THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 7: **Measurement of Time and Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The formula for speed is: 1

(a) distance / time **(b)** distance + time **(c)** time / distance **(d)** distance x time

2. In a distance-time graph of a uniform motion, the graph is a: 1

(a) circle **(b)** zigzag **(c)** straight line **(d)** curve

3. The time period of a simple pendulum depends on its: 1

(a) amplitude of small swings **(b)** material of bob

(c) mass of bob **(d)** length

4. A car travels 120 km in 2 hours. Its speed is: 1

(a) 30 km/h **(b)** 60 km/h **(c)** 240 km/h **(d)** 120 km/h

5. A horizontal line in a distance-time graph indicates the object is: 1

(a) accelerating **(b)** at rest **(c)** moving uniformly **(d)** falling

6. If a pendulum makes 20 oscillations in 40 s, its time period is: 1

(a) 40 s **(b)** 0.5 s **(c)** 2 s **(d)** 20 s

7. A cyclist travels 100 m in 20 s. Its speed is: 1

(a) 2000 m/s **(b)** 0.2 m/s **(c)** 5 m/s **(d)** 20 m/s

8. A steeper straight line in a distance-time graph means: 1

(a) lower speed **(b)** rest **(c)** zero distance **(d)** higher speed

9. A sundial measures time using: 1

(a) sand **(b)** a pendulum

(c) water **(d)** the shadow of a rod

10. A train moving at 60 km/h travels in 3 h a distance of: 1

(a) 20 km **(b)** 63 km **(c)** 57 km **(d)** 180 km

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Convert 54 km/h into m/s. 2

12. Name two ancient time-measuring devices. 2

13. What is an oscillation? 2

14. What is a distance-time graph? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe an activity to find the time period of a simple pendulum. 3

16. Explain how a sundial works and why it fails at night. 3

17. A stone falls and covers 5 m in 1 s, 15 m in 2 s. Is motion uniform? 3

OR

Explain why we measure the time for 20 oscillations rather than for 1. 3

18. Describe the working of a water clock. 3

OR

How does the time period of a pendulum change if its length is increased? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the development of time-measuring devices from the sundial to the quartz watch. 5

OR

Describe the measurement of time using a stopwatch, with precautions. 5

20. Describe speed, uniform and non-uniform motion with examples and the distance-time graph for each. 5

OR

Explain the importance of standard units of time and how we use hours, minutes, seconds. 5

END OF QUESTION PAPER — Total: 40 Marks

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